DANIEL KOCH

SOFTWARE DEVELOPER

SUMMARY

Software developer specializing in real-time video and low-latency streaming for life-critical medical applications. Skilled in WebRTC, GStreamer, mediasoup, NDI, Python, OpenCV and C++, with full-stack proficiency in TypeScript, Node.js, and React.

CONTACT DETAILS

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EXPERIENCE

VIDEO DEVELOPER

Nortal - Caresyntax | 2020 - present

- Developed a remote surgical visualization platform for cloud based viewers using WebRTC SFU with mediasoup, React, and Node.js
- Designed and implemented GPU-accelerated NDI camera streaming services in C++, GStreamer, NDI SDK, and Python, utilizing NDI HX with H.264, VP8, and Intel VAAPI codecs
- Built surgical recording services in C++, GStreamer, NDI SDK, V4L2, and Python, leveraging NDI HX (HEVC/H.264) and hardware encoders such as Magewell AIO for high-quality, low-latency video capture

FULL STACK REACT DEVELOPER

Solution for Life| 2019 - 2020 (full-time)

- Developed SFL.Money, a personal finance management platform, as a solo full-stack developer
- Built front-end in React and back-end in Node.js and TypeScript

C / FULLSTACK DEVELOPER

Blockbit| 2015 - 2019 (full-time)

- Developed back-end services, network scanners, and protocols for BLOCKBIT VCM, a Vulnerability and Compliance Management tool
- Expanded into full-stack development, building front-end interfaces in React and Meteor alongside back-end services in C and Node.js

COMPUTER VISION DEVELOPER

3RCorp | 2008 - 2015 (full-time), 2015 - current (contract)

- Developed PeopleFlow (pedestrian traffic tracker), Penalty D-Tech (volleyball ball tracking), and Itaú Vault (Kinect-based bank vault access control)
- Integrated WebRTC-based remote monitoring using headless Chromium in Electron with Xvfb under systemd